Recovering from a bounded set of images

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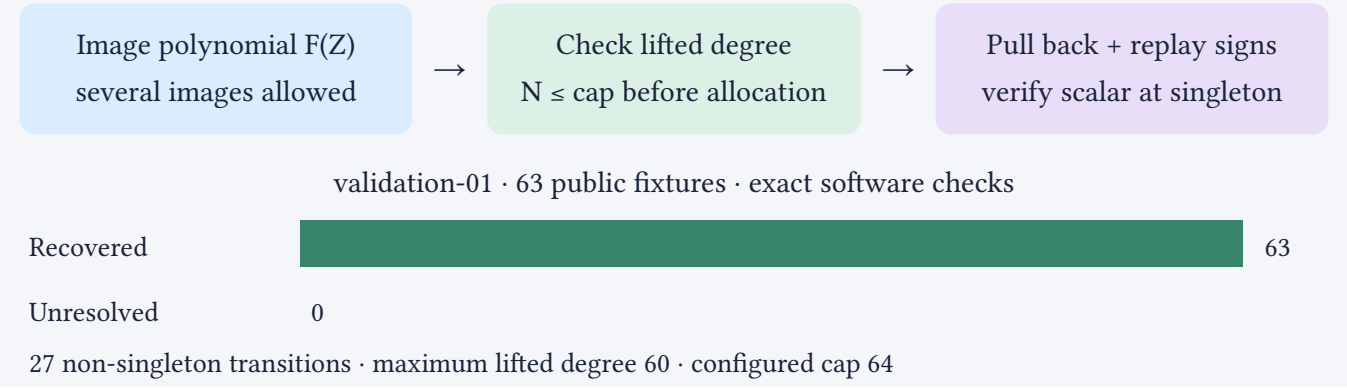
Weighted modes can now transition to scalar recovery before the image becomes unique. A capped polynomial pullback preserves every candidate fiber; replaying already collected signs refines it without new source queries.

For a monic divisor $F(Z)$ of $Z\left(Z^d-1\right)$, let $\varepsilon=1$ when $F(0)=0$, and zero otherwise. Then

$$L(X)=\frac{F(X^m)}{X^{(m-1)\varepsilon}}$$

is squarefree and has degree $N=m\deg(F)-(m-1)\varepsilon$. Its roots are exactly the scalar preimages of the retained images. The zero correction is essential. Check N against the cap before allocating.



Public results. All63 inherited fixtures recover, including the17 F509/r479/m2 cases unresolved by the preceding four-radius mode-only adapter. The new path records511 source calls,180 retained-sign replays and115 extra queries. Every candidate passes public-point verification. Different control flows and charged replay work prevent interpreting source counts as a timing comparison.

Finite checks. 712 image subsets;356 containing zero;2,756 replay prefixes;689 cap rejections. Exact root sets and degrees agree with an independent enumeration reference. All63 public fixtures also exercise the one-call stopping control. Reference enumeration is confined to checks.

Bounds. Four prescribed radii; scalar degree cap64; at most17 prescribed extra offsets; source-call cap128. Image/compiler memory and source setup are separately charged. Replay uses modular character powers and gcds at the shrinking lifted degree; no scalar or image root list is built at runtime.

Evidence and scope. validation-01, modes.rs, pullback.rs and generation.json preserve source, hashes, raw outputs and warnings. The unchanged prior adapter still reports its17 unresolved cases. Prime-field source restrictions and the original4,563-case cohort remain unchanged. No elapsed-time study, full formal proof or independent signed verdict is supplied.

Next: reduce repeated modular-power work during retained-sign replay, while preserving the exact set semantics and fixed cost accounting. Source batching and domain extension remain separate construction tasks.

Sources: local weighted-mode, retained-fiber and half-GCD components; FLINT nmod_poly conventions (flintlib.org/doc/nmod_poly/index.html). KB unavailable; fallback recorded in DERIVATION.md. Polynomial inflation and gcd are standard operations.